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REINFORCEMENT LEARNING

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UNIT 2

DYNAMIC PROGRAMMING AND MARKOV DECISION PROCESSES

STUDY GOALS



- Define and structure **sequential decisions problems**
- Understand and apply optimality rules for effective **decision-making in sequential processes**
- Learn how to implement **optimization principles** through computer programs and applications
- Explore the significance of **utility theory** in various applications

EXPLAIN SIMPLY



1. Explain the concept of dynamic programming and its practical applications in sequential decision making

2. Describe the key components of Markov Decision Processes, including policies, value functions, and Bellman's Equation

3. Discuss the role of value iteration and policy iteration methods, in addition to utility functions in optimizing decision-making processes

APPLICATIONS OF DYNAMIC PROGRAMMING

OPERATIONS RESEARCH

Resource allocation optimization, scheduling

ECONOMICS

Optimal investment and savings strategies

DYNAMIC PROGRAMMING

ARTIFICIAL INTELLIGENCE

Decision-making, problem-solving, game playing

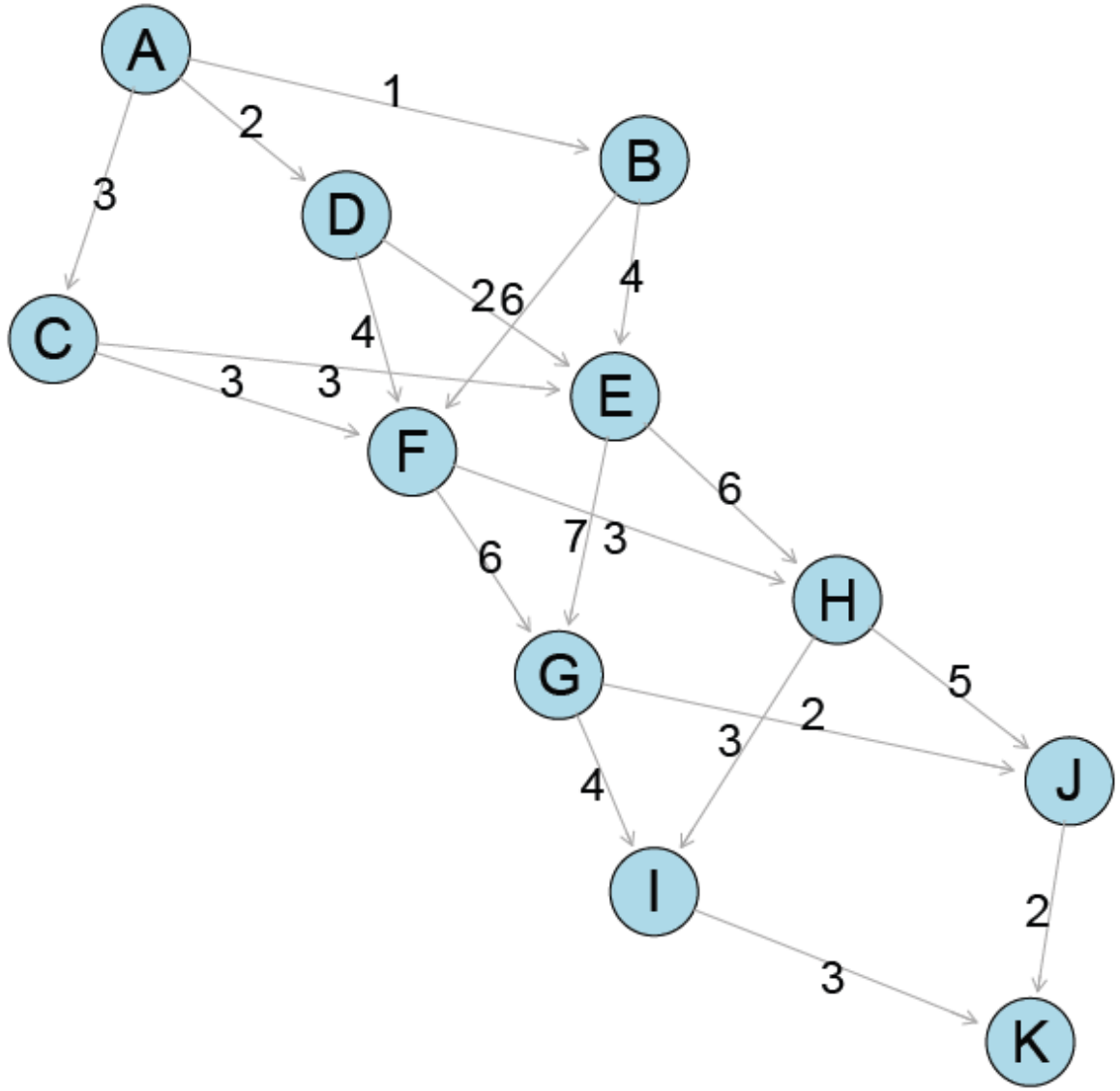
ROBOTICS

Autonomous vehicles, path planning, robot control

Source of the text: Adda, Cooper, & Cooper 2003; Sutton and Barto, 2018; Xue, Li, & Yang, 2019.

Stagecoach problem: find the route with minimum total distance

- **Nodes**: labeled towns (A - K)
- **Edges**: routes between tows
- **Labels:** distance values on directed edges





Strategy?

Source of the image: Own representation.

Enumeration Strategy

Exhaustively list, evaluate, and compare all possible routes

- Count all possible routes
- Calculate total distance for each route
- Choose shortest route



Time increases exponentially

$$N_{route} = n_1 \cdot n_2 \cdot \ldots \cdot n_k$$

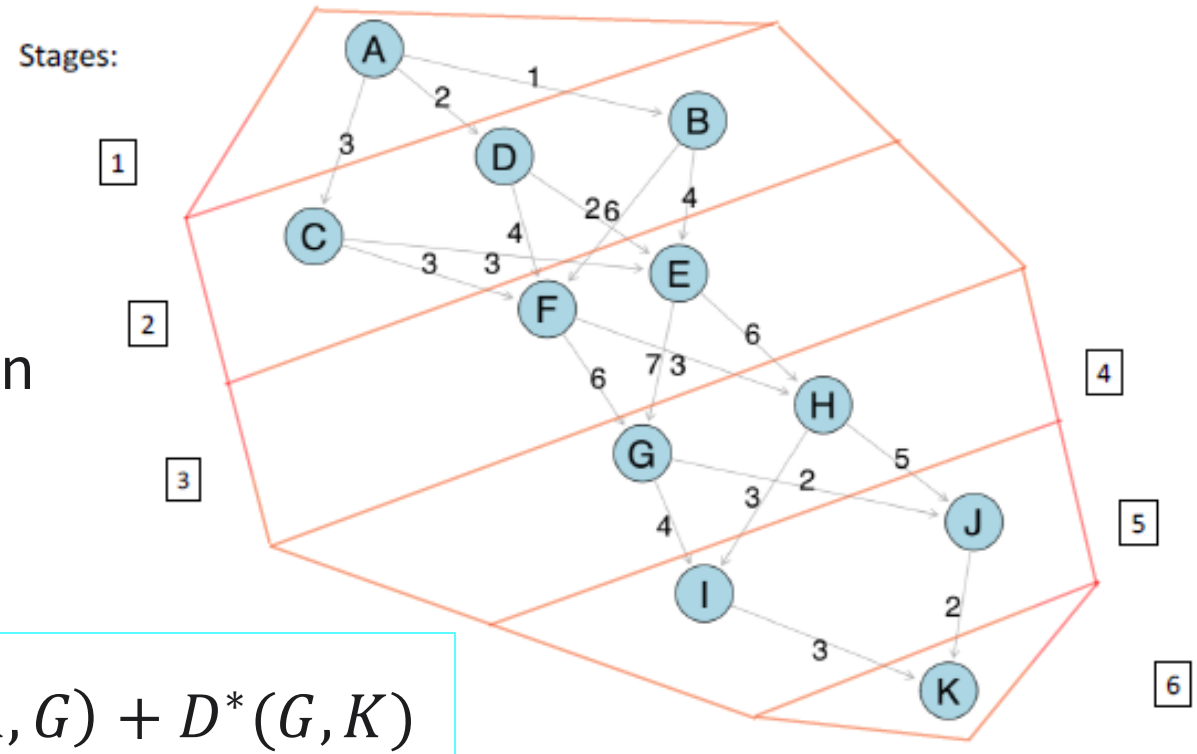


Can complexity be reduced?

Divide and Conquer Strategy

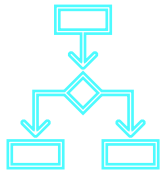
Decompose into independent sub-problems for efficient optimization

- Decompose into route segments
- Minimize segments independently
- Consider minimum distance between nodes in distinct stages



$$D^*(A, K \mid G) = D^*(A, G) + D^*(G, K)$$

Source of the image: Own representation.



Sequential Decision Process (SDP)

- **Stace space:** set of possible states
- **Action space:** set of available actions
- **State-action space:** combined states and actions
- **Cost function:** associated costs for actions in states
- **Transition function:** model state changes



SDP comprises stages with costs and actions, shaping future outcomes via state transitions and decision-making

Source of the text: Bellman, 1957.

COMPONENTS OF THE DYNAMIC PROGRAMMING MODEL

Memoryless SDP

- Behavior depends solely on current state-action
- Transition, cost, and decision rules follow the same principle

SDP Lifetime

- Fixed/finite, infinite, or variable stages
- May include a terminal stage

Time Homogeneity

- Changing components → time inhomogeneous
- Constant components → time homogeneous

SDP Cost Accrual

- Stages' costs combine into single value

$$R = R(x_1, a_1) + \cdots + R(x_n, a_n)$$

Discounted Cost Accrual

- Accrues costs with future values discounted

$$R = \Sigma_{j=1}^{\mathrm{N}} \beta^{j-1} R(x_j, a_j)$$

MiniMax Cost Accrual

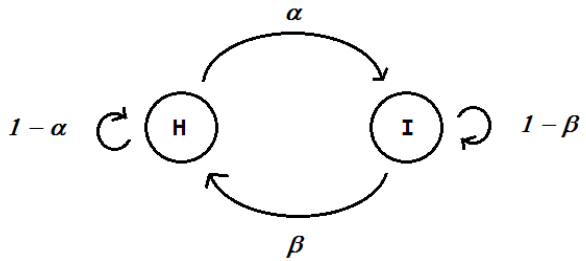
- An SDP is a minimax cost problem

$$R = \max\{R(x_1, a_1) \ldots R(x_n, a_n)\}$$

Source of the text: Sutton and Barto, 2018.

Markov Decision Process (K, R, P, β)

- State action space K
- Cost function $R(x, a)$
- Probability kernel $P(y \mid x, a)$
- Discount factor β



Bellman Principle of Optimality

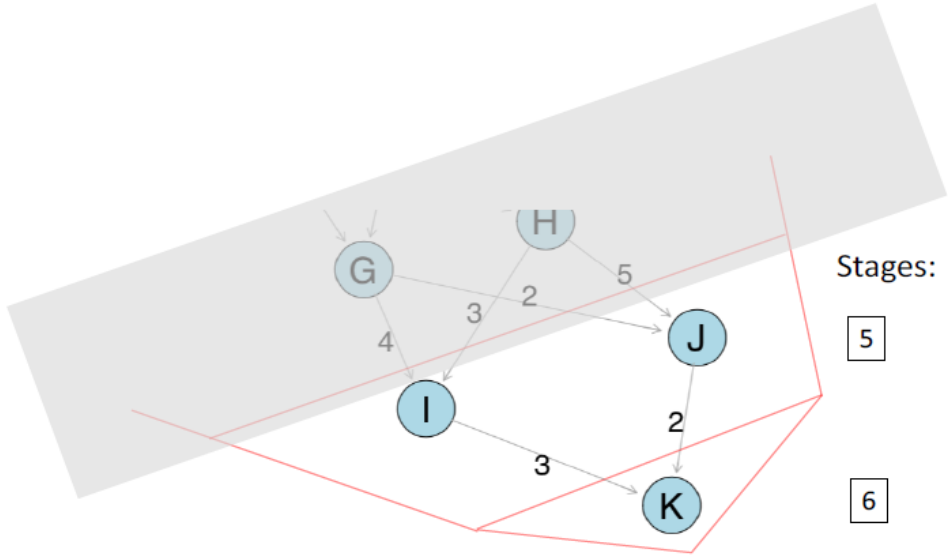
- Find and evaluate a policy function to optimize cost accrued
- Current decisions affect future optimality

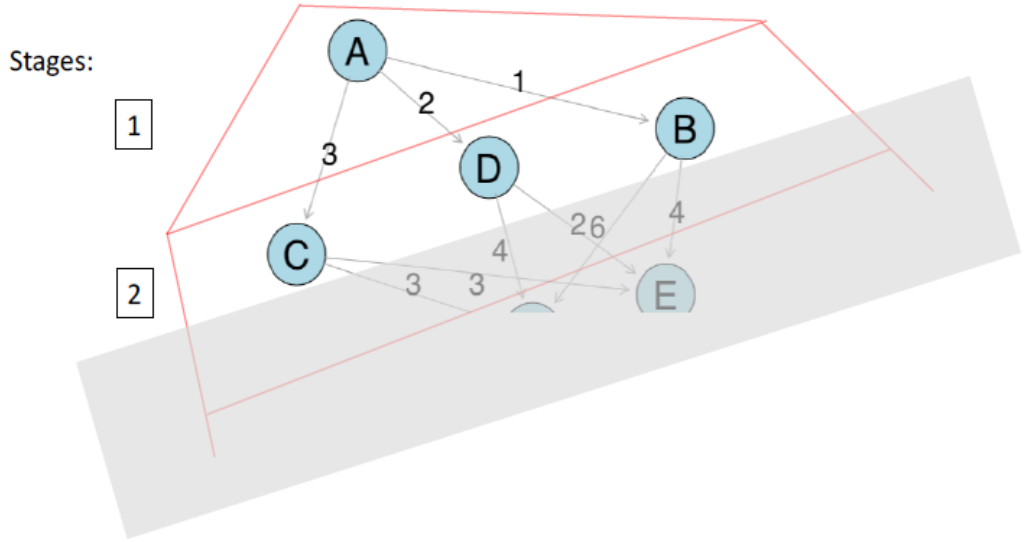
$$V^*(x) = \min_{a \in X} V^*(x, a) \qquad x \in X$$

Source of the text: Sutton and Barto, 2018.
Source of the image: Own representation.

Backward induction approach for stagecoach problem

- Determine optimal policy for final stage
- Backward policies calculated, utilizing terminal state
- Sequentially compute policies for preceding stages.





Source of the image: Own representation.

Value Iteration
Iterative Optimization

- Initialize values for each state
- Update values using Bellman's equation
- Calculate new values based on maximal reward
- Converge to optimal value function

Policy Iteration
The Actor – Critic Method

- Policy evaluation **(Critic)**: evaluate V^{Φ_i}
- Policy improvement **(Actor)**

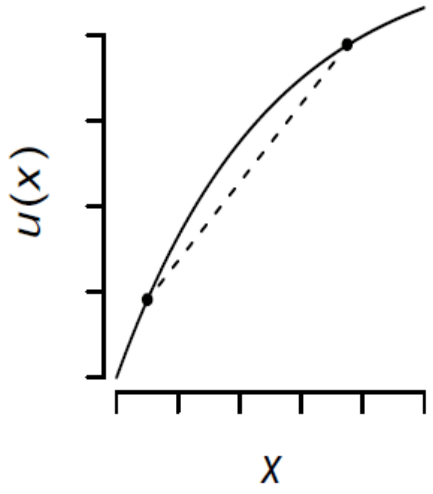
$$\phi_{i+1}(x) = \underset{a \in A_x}{\operatorname{argmin}} R(x,a) + \beta \sum_{y \in X} V^{\phi_i}(y) P(y \mid x, a)$$

- Terminate when optimal policy is found

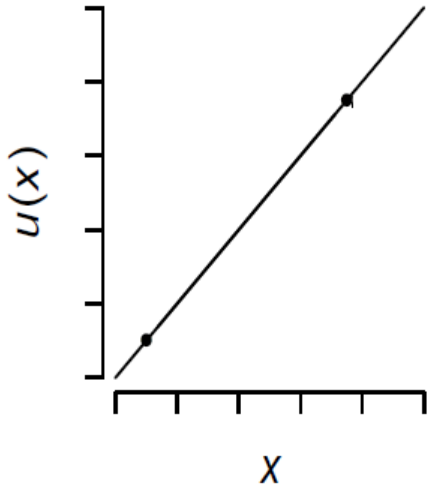
Source of the text: Sutton and Barto, 2018.

Utility Functions

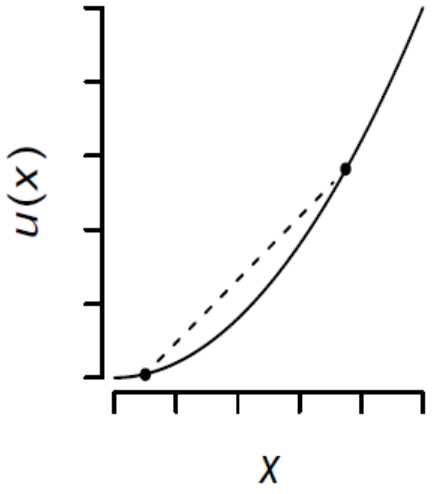
Quantify preferences (cost, reward, risk) for diverse decision makers



Risk Averse(concave)



Risk Neutral (linear)



Risk Seeking (convex)

Source of the image: Own representation.

REVIEW STUDY GOALS



- Define and structure **sequential decisions problems**
- Understand and apply **optimal rules** for effective decision-making in sequential processes
- Learn how to implement **optimization principles** practically through computer programs and applications
- Explore the significance of **utility theory** in various applications

SESSION 2

TRANSFER TASK



Case study



A navigation app aims to enhance travel efficiency by finding the shortest route while minimizing travel time and fuel costs. The challenge is to identify the optimal route, considering variables like road distance, traffic conditions, vehicle fuel efficiency, and user satisfaction

Task

Discuss how you can utilize MDP for route planning. Define states, actions, rewards, transitions. Optimize route selection

TRANSFER TASK
PRESENTATION OF THE RESULTS

Please present your results.

The results will be discussed in plenary.





1. A traveling salesman needs to visit 10 cities. A tour starts at a home city x = 1, visits each city exactly once, then returns to x[1]. The fifth city visited must be a specific city x = 5. The object is to determine which tour is the shortest, in total kilometers, under the given constraint. If the problem was to be solved by complete enumeration, how many tours would need to be evaluated?
   a) 5,040
   b) 14,400
   c) 40,320
   d) 362,880



2. The total reward for a two-stage sequential decision process is given by R = u(x[1]) + u(x[2]), where u(x) = sqrt(x) is a utility function; and x[1] and x[2] are nonnegative, and constrained to satisfy x[1]+x[2] = 50. What is the maximum total reward?
   a) 2 / 5
   b) 10
   c) sqrt(50)
   d) 2 * sqrt(50)



3. A simple game consists of three stages. Each stage k = 1,2,3 has a reward R(k), say R(1) = 20, R(2) = 20, R(3) = 30. Each stage k also has a quitting cost C(k), say C(1) = 5, C(2) = 35, C(3) = 60. At stage k, the player earns the current stage reward R(k), then decides whether or not to quit. If the player quits, he keeps all rewards earned so far, but deducts from that total the stage k quitting cost C(k). The game ends at that point. The player starts at Stage 1 and must quit by the end of Stage 3. What policy maximizes the player's total reward (net total cost)
   a) The optimal policy is to quit at Stage 1
   b) The optimal policy is to quit at Stage 2
   c) The optimal policy is to quit at Stage 3
   d) All policies yield the same reward

LIST OF SOURCES

Adda, J., Cooper, R., & Cooper, R. W. (2003). Dynamic economics: Quantitative methods and applications. MIT Press.

Bellman, R. (1957). Dynamic programming. Princeton University Press

Sutton, R. S., & Barto, A. G. (2018a). Reinforcement learning: An introduction (2nd ed.). MIT press.

Xue, S., Li, Z., & Yang, L. (2019). Training a model-free reinforcement learning controller for a 3-degree-of-freedom helicopter under multiple constraints. Measurement and Control, 0020294019847711

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